Investigating the Leverage Effect on **the Polish Stock Market** Using Principal Regression Analysis

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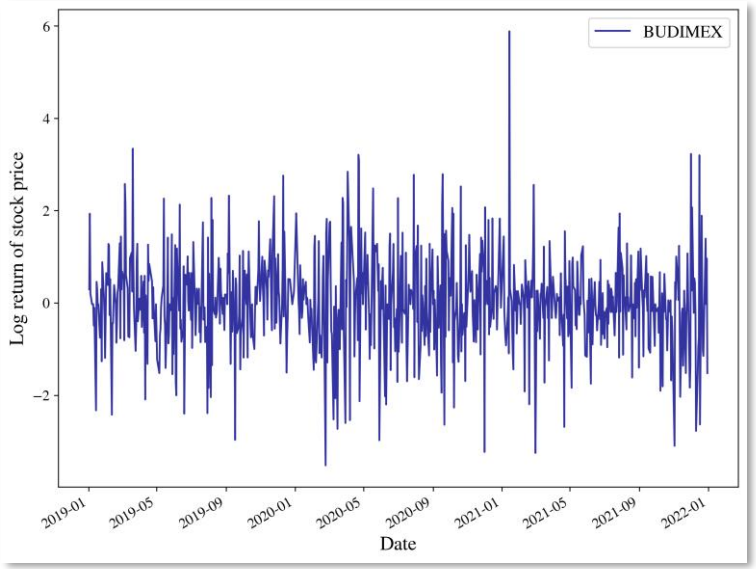
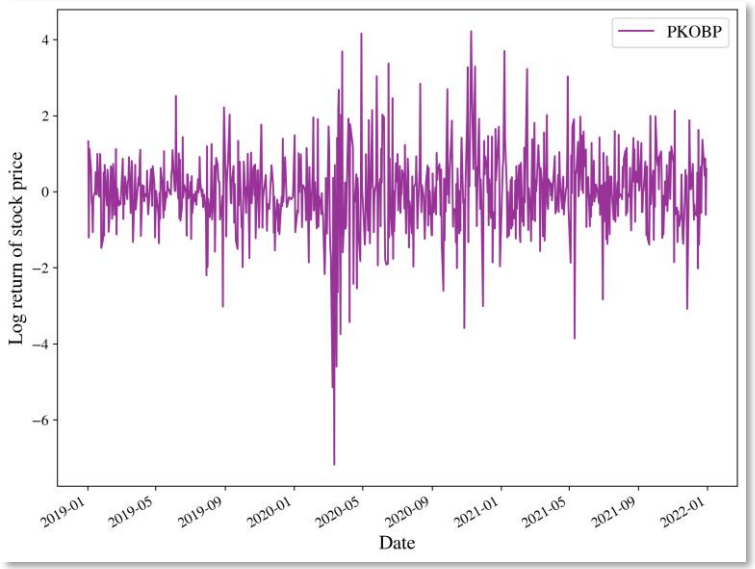
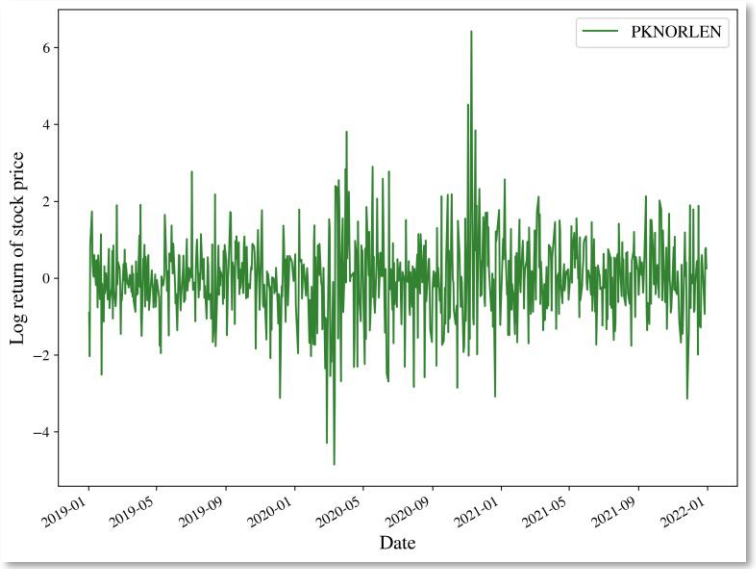


$$r_\alpha(t) = \log \frac{p_\alpha(t)}{p_\alpha(t-1)}$$





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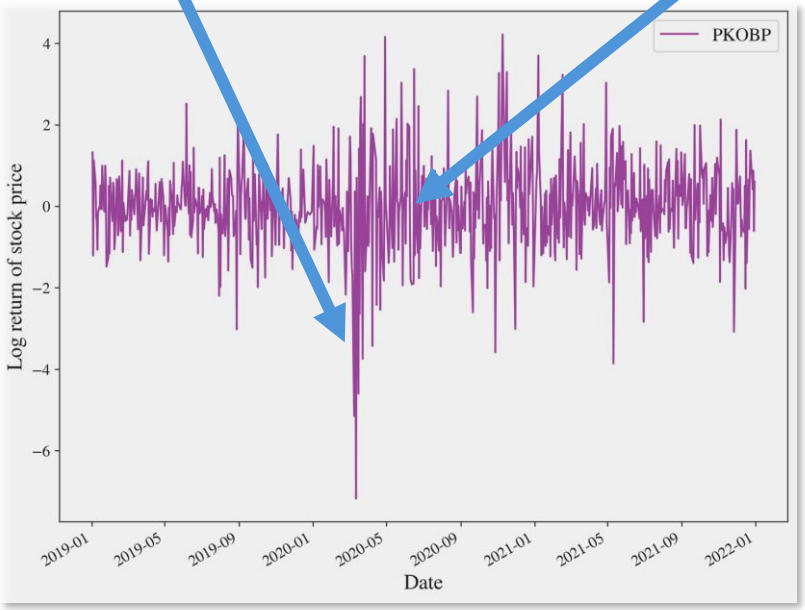
Negative returns

Increased fluctuations



The Index Leverage Effect

Fluctuations of an Index

- Fluctuations of stocks
- Correlations between stocks

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$$r_\alpha(t) r_\beta(t) = \mathrm{C}_{\alpha\beta} + \mathrm{D}_{\alpha\beta}(\tau) I(t-\tau) + \varepsilon(t,\tau)$$

$$\Downarrow$$

$$\mathbf{C}(I) = \mathbf{C_0} + I\mathbf{D}$$

$$\Downarrow$$

Dynamics of the stock market:

Eigenvalue decomposition of $\mathbf{C}(I)$